Visualizing Modeled Social Networks: a Provocation*

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Abstract: Network scholarship regularly leverages visualizations. They are thought to enhance readers' recall, interpretation, and simplification of scholar's intended takeaways. However, by representing the specifics of single empirical cases, these visualization strategies may not optimize how well they accomplish our scientific purposes. In particular, they may: (1) be overly complex in ways that obfuscate what we can learn from them, (2) draw attention to the particularities of the case, rather than the general principles the case is examined to characterize, (3) be difficult to precisely replicate, raising questions about consistency in interpretation, and (4) introduce potential risks that could violate research ethical standards. To address these, this paper provokes the field to consider whether presenting *modeled* network results–in lieu of empirical cases–retains the aims of network visualization, while potentially overcoming some of the limitations of existing practice.

NOTE: This is very much a fledgling idea that isn't yet ready for citation/sharing. If you're interested in helping me bring this to life, please reach out. Otherwise, for now, think of it as a musing... Also, I'd initially envisioned submitting as a Viz paper to *Socius*, and have since submitted it to *Social Networks* for their theory special issue (which has a submission timeline of Nov 1.)

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Social network research relies heavily on data visualization (Freeman 2004). These visualizations can serve to characterize, simplify, and focus theoretical or empirical claims (Moody et al. 2005; Rawlings et al. 2023). Visualizations are generally more readily digestible than tables of seemingly endless coefficients, significance tests, and summary statistics (Healy 2018; Schwabish 2021). Additionally, with the increasing ubiquity of network verbiage in society, they can even convey ideas in ways that require less "translation" work for audiences who sit outside of academic and research niches (Sayama et al. 2016; Wolff et al. 2026). Quite simply, network visualization is not only common, but can be incredibly useful (Birkett et al. 2021).[1]

Traditionally, this has taken the form of directly visualizing instances(s) of empirically **observed** social networks–whether snapshots of single data waves, or evolving network "movies" for dynamic relational data patterns (Ognyanova 2025).[2] That is, such visualizations are intended to characterize the data *per se*. While this practice predominates, it's worth asking: (1) why we visualize, and (2) what optimal set of principles govern strategies for accomplishing those aims, then (3) whether an alternative strategy serves those *purposes* and *needs* better than the existing approach?

In this paper, my answers to this set of questions suggests many practices of social network visualization are currently misaligned to their purposes–or more precisely, assumes a single visualization orientation can successfully address multiple (sets of) aims. As is often the case with split allegiances, I contend the predominant strategy leads us to occasionally miss the mark, especially when attempting to convey the *theoretical* takeaways of social networks research.

To address this misalignment, I propose that many (though notably, not all) purposes of social network visualizations would be better served by visualizing modeled networks rather than observed network data. After making this case following the motivations above, I illustrate this shift in three ways–a trivial example common to how many of us already teach social network modeling, describing a parallel strategy common in non-networks research, and an applied example from a recent research paper.

Before continuing, I should emphasize that when I say "model" I intend this usage in a *theoretical*, not a *statistical* sense. I avoid relegating this point to a footnote to highlight that the strategy I encourage does not apply only to statistically modeled results. Moreover, I should emphasize that this broader applicability in some ways reflects that my proposal in places *imports* practices from non-statistical orientations to network research as much as (or perhaps even more so) than developing a method for statistical applications that can be exported to other use cases. That caveat noted, my illustrations below will primarily draw on statistical applications as a concrete demonstration; but I will include how the *principles* demonstrated are applicable beyond these particularities.[3]

I need to better flesh out the purposes/principles distinction here and ensure I am convinced I have them in the right buckets.

Purposes & principles of (network) visualization

So, what are the benefits of network data visualizations? And what are the principles we should follow to best accomplish those aims? Here, I begin from the assumption that SNA visualizations seek to optimize their *scientific utility*, and highlight some shortcomings that have emerged in our common practices, which have weakened our ability to accomplish those aims–particularly by assuming single visualization strategies can address multiple (sometimes competing) sets of aims.

Many of the benefits for network scholarship mirror those that motivate data visualization in science more broadly (Rawlings et al. 2023). A fundamental premise is that data visualization enhances **memory** (Schwabish 2021); this reflects what we know from a number of old adages–including that there's simply value

[1]As with any data presentation strategy–visual or otherwise–there are also ample examples of how to do this poorly (e.g., what have come to be known as unintelligible network "hairballs"); I have a few published visuals of my own that I'd *substantially* rework if given the opportunity today.

[2]These typically occur *after* data pre-processing has taken place (e.g., cleaning, recoding, etc.), but no further data reduction steps (i.e., are not the product of data *summaries* or analytic interpretations).

[3]Specifically, some of the elements proposed below are directly adopted from other corners of SNA where they are already variably common practices (if labeled differently), and (partially) pave the way for how statistically oriented researchers can implement the approach.

in beauty (Tufte 2006; Halpern 2015). Moreover, this memory benefit leverages the potential **efficiency** available in data visualization that simply cannot be achieved in other strategies for presenting evidence (Healy 2018). Edward Tufte has prioritized not only using this benefit, but attempting to find ways to optimize it, e.g., via efforts to maximize what he conceptualized as the "ink-to-information" ratio (Tufte 1983). These memory-oriented enhancements can arise both via improved recall and encoding processes–on the latter, the simple act of including visualizations can enhance memory via the complementary of multimodal communication strategies, dovetailing with the strengths and weaknesses of text (Sweller 1988).

The data we use to represent social processes are almost always inherently high-dimensional, and visualizations provide a means for *simplifying* that complexity in structured, interpretable formats (Wickham et al. 2023).[4] This leads visualizations to not only enhance communicative aims, but also to improve readers' ability to understand and interpret complex social phenomenon (Schwabish 2021). While statistics can summarize data in principled ways, data visualization rarely simply mimics these other forms and representations of data reduction. Instead, they are also "attention-drawing" efforts that highlight certain features of the data, while typically necessitating the deprioritization of others (Leifer 1992).

So, how do network scholars *in particular* prioritize these combating aims? Network scholars especially leverage these dimension reduction approaches by translating summaries of (theoretically) salient social space into the particular strategies for visualization forms (Rawlings et al. 2023). For example, one such "attention-drawing" move researchers engage in can seek to highlight certain elements of (theoretically salient) relational patterns (e.g., community structure), while deprioritizing others (e.g., homophily).[5]

While such trade-offs may seem universally useful, I argue that how researchers prioritize among these various aims should differ according to different stages the research process when we're leveraging visualization as a tool. Moreover, I will contend that collapsing the "preferred strategy" across these non-overlapping sets of aims has been one source of limiting the utility of SNA researchers' visualizations. That is, my hope is the encourage, more, more-varied, particularization uses of visualizations in ways that will afford each instance greater likelihood of achieving the researchers' purposes for visualizing their network(s). In particular, optimal visualization strategies differ substantially between exploratory (Wickham et al. 2023), conceptual/theoretical (Brady et al. 2020), or explanatory aims (Healy and Moody 2014).

Differentiated aims according to research stage

Given the potential multiplicity of visualization use cases' aims, it is worth considering how readily our current approaches address the prioritization among aims. First, if data fidelity, anomalous cases, and idea generation are of particular interest in exploratory analyses, that would prioritize a set of *data* features as the key elements for visualization to highlight. As such, exploratory network visualizations can serve to characterize a study population's composition and structural patterns, in ways that help to articulate things like: features that analyses seek to explain, or identify needed pre-processing steps (Wickham et al. 2023).

Second, for conceptual development in formalizing the aims of a research question, visualization likely needs to optimize on elements of the *analytic strategy* - e.g., differentiating which components are explicitly examined, versus which are (unexamined) assumptions. Conceptually, network visualizations necessitate a concrete articulation of what analyses will and will not explicitly address, highlighting aspects of a theorized social and relational process to be evaluated (adams 2020).

Third, if being used as a strategy for presenting analytic *results*, the emphasis typically seeks to highlight *theoretical* implications of interpretation. In social network research with explanatory (even causal) aims, researchers seek to demonstrate how the observations in empirical cases (Kadushin 2012), and appropriate analytic strategies conform to expectations in ways that (fail to) allow appropriate theoretical interpretations (Borgatti and Lopez-Kidwell 2011). In this case, visualizations can prioritize that *conformity* over "data

[4] Some contend that this is a *constraint* of data visualization imposed on data presentation strategies (by obfuscating its richness), rather than its benefit (Healy and Moody 2014).

[5] For example, to emphasize the changes in political polarization over time in the US Congress, Moody and Mucha (2013) employ increased coarse-graining to highlight *group* size and positioning over time, and juxtapose that against the decreased *individual* positioning and distancing.

fidelity" or theoretical "ideal types" alone, and likely warrants different optimizations to clearly convey those implications (Bender-deMoll and McFarland 2005).

The above leads to the implication that while all visualization prioritizations and resulting strategies necessarily employ **simplifications**, the potential non-overlaps among those (prioritized) aims could lead to sub-optimal visualizations in a number of distinct ways.

Perhaps most obviously, when researchers construct a visualization to serve one of these purposes (e.g., exploratory data analysis), it will not optimally serve other aims (e.g., explanatory), because the simpllifications necessary for one purpose mask what needs to be optimized for another. The previous discussion elaborates why we should reject this assumption; whether we have been making it implicitly or intentionally.

Perhaps more problematically, when researchers *assume* (or worse, *require*) visualizations have "universal" benefits, we are presupposing that the purposes of data visualization are necessarily shared. One common version of this approach in SNA visuals has been to produce a "descriptive" plot of the observed data, then to "overlay" some features of the theoretical interpretation of analytic results in hopes that readers can be taught to see the "important" signal among the other noise in the data). Unfortunately, in this case, we are likely to find ourselves in a situations where the visualizations strategies we employ can end up failing to accomplish *any* of our aims well (Pache and Santos 2010). My contention is that network visualizations are not infrequently commiting one of these two errors.[6]

As such, in the remainder of this paper, I elaborate explore two distinct aims. The first path develops a proposed mapping of differentiated prioritizations to optimized visualization strategies. The second path seeks to demonstrate how "modeled network visualizations" offer a strategy for conveying the theoretical implications of SNA empirical analyses in ways that are presently under-utilized in the field.

Additional principles for optimal (network) science visualization

Before jumping into those tasks for optimizing across multiple primary scientific purposes in visualizations, I also need to address two other desirable features of (network) data visualization, that any approaches should incorporate.

While the premised benefits of open science principles are not new (Partha and David 1994), researchers have recently (re-)discovered enthusiasm of better implementing these aims in practice (Miguel et al. 2019).[7] Visualizations themselves can raise thorny questions about replication. Network visualizations entail a combination of node/tie inclusion/exclusion rules, their respective attribute information, and determinations about positioning for each of these.[8]. A first consideration therefore is what scope (which variables for which nodes and which ties) and granularity of information (e.g., levels of measurements per observation) are included in the visualization; these *may* need not be full copies of their presence in the raw data. Furthermore, given that *positions* in network visualizations are typically defined relatively (rather than absolutely), algorithmic and procedural details (e.g., random number generator seeds) can be important details for faithfully reproducing visualizations that are substantively, or even aesthetically comparable (Bender-deMoll and McFarland 2005).

Additionally, network visualizations often present unique ethical challenges in how we share our research (adams 2019; Ortiz Ruiz et al. 2026). Especially in "closed population" settings (but see also Zimmer 2010), the potential for deductive disclosure is broad in any presentation of multiple variables (Rocher et al. 2019). Relational data provide additional avenues for (serial revelations) of deductive disclosure (Narayanan and

[6] For example, in instances where I've wanted to convey the theoretical takeaways from in adams (2013) –that different types of "risk partnerships" provide different contributions to overall graph connectivity, even while their node-level differences/combinations are minimal–I've found that nearly impossible to readily convey from the (largely exploratory-oriented) visualization included in the paper–error 1.

[7] In fact, in personal communication, one colleague noted that the "sociogram is. . . one of the purest original forms of open science, [. . . and] replicable data transparency."

[8] Given the aims of this paper, I focus on the replicability of *visualizations* independently of the numerous additional appropriate considerations for how to implement reproducible research more broadly (Miguel et al. 2019), including for the particular demands of network research (Neal et al. 2024; Pasquale et al. 2025)

Shmatikov 2009).[9] To this end, our frequent practice of plotting empirical networks from data collection efforts may undermine principles of confidentiality and minimal risk in ways that should lead us to consider alternate visualization strategies that can simultaneously leverage these strengths while minimizing these potential risks (Ortiz Ruiz et al. 2026). These potential limitations do not preclude network researchers from being able to visualize, and even share our data (Pasquale et al. 2025); instead suggesting that we likely need to tailor our approaches to doing so in ways that current practices don't necessarily optimize.

These considerations of (1) multiple potential scientific aims, (2) desire for replicability, and (3) reducing ethical harms, can each lead to different optimization strategies for visualizing social networks in research. Moreover, their combination likely prioritizes different strategies than any one of them alone may lead us to conclude as the best visualization strategy to employ. In the following section, I describe a stylized typology of visualization strategies, and consider the trade-offs that apply across these strategies in terms of their ability to accommodate this set of principles. To foreshadow, I argue that different strategies are optimal for different tasks, describe a use-case that is a particular combination of these aims, then ilustrate a presently under-used strategy for visualizing *modeled analytic results* of networks to better satisfy that combination of aims.

How & what do we visualize?

The "traditional" approach

An overwhelmingly common orientation to visualizing network data takes an approach best understood as it simply representing one way of conveying information (about the nature of the data at hand). In this orientation, visualizations provide a first step towards presenting one's case. "First step" however doesn't mean that the data presented are "raw." It merely indicates that the data have been obtained, minimally processed, shaped into a coherent presentation strategy, and presented to the reader. As noted in the discussion of replicability above, each of these steps requires implementing (variably justifiable) decisions, that shape the nature of the visualization, but our (implicit) assertion is typically that what we present in these types of visualizations are merely "representing" our case.

The fundamental steps to generating such "observed" networks entails: (1) obtaining the data (through primary methods or secondary sources), (2) pre-processing, cleaning, and organizing the data, (3) reducing to focal elements (relational and nodal attributes), then (4) transforming that reduction to visual form, which we then (5) label. The result is a representation of the "observed" network that is to be analysed (i.e., highlighting–parts of–what the research aims to address, while necessarily excluding other elements). This type of visualization serves a similar purpose to the "descriptive statistics" Table 1 that many papers provide - giving the reader a means to orient to the case at hand with a distilled summary.

Conceptual visualization(s)

Before moving on to describing the approach I'm encouraging here, we should also address a related consideration that is not often equated with visualization, because it is typically done without data. Some fields readily draw on *conceptual models*, to varying degrees of formalization–ranging from those that primarily serve to "organize one's thoughts" (Brady et al. 2020) to approaches that formally articulate analytic aims and assumptions (e.g., directed acyclic graphs Elwert 2013). Similarly, network research has leveraged these types of visualization aims to specify how *combinations* of features can lead expectations to diverge from various "ideal type" formations of networks–doing so in concrete ways that may not be as readily intelligible without visualizations.[10] Conceptual visualizations can be used to serve dual purposes–complementary formalizing researchers' expectations of what revealed (combinations of) patterns would align with specific (hypothesized) interpretations, while also articulating what revealed "exclusions" would lead to disconfirming or alternate

[9] I.e., the revealed identity of one node in a graph (e.g., myself) can in turn heighten the capacity to identify others' identities in turn (e.g., my friends, etc.)

[10] A very widely used example of this is how "small world" networks, while being very differentiated from random networks, require few shortcuts, which does not make them appear particularly statistically distinct from "caveman" graphs, but the difference is readily apparent when visualized (Watts and Strogatz 1998).

interpretations. That is, conceptual visualizations serve to illustrate the theoretical claims being evaluated in research.[11]

Modeled network visualizations

My fundamental proposition is simple - instead of plotting the "observed networks" directly in the "traditional" approach, we can instead produce visualized networks that characterize analytic results in ways that may better achieve certain combinations of the aims and principles outlined above. In a number of ways, it is not a ground-breaking suggestion. In fact, it's a practice many statistical modelers–of social network, and other types of data–already enact in practice; but we don't deploy those approaches in publications in ways that optimally leverage their utility. For example, "eye-balling" consistency of modeled results with observed networks–in tandem with formal statistical tests for goodness of fit–is even typically taught as a helpful practice in model development and fitting processes (Krivitsky et al. 2023). However, while we use these heuristic network visualizations to improve model fit (whether for capturing theoretical limitations, or for enhancing stability of estimates), when presenting networks in publications, authors still often resort to presenting the observed version as our primary network visualization(s). I've done this multiple times myself.

Table 1: Potential Comparative Benefits across Visualization Strategies

purpose or principle	observed	conceptual	modeled
explore/describe	✓	x	x
simplify	?	✓	✓
reproduce	?	✓	✓
confidentiality	x	✓	✓
theorization	x	✓	?
empirical interpretation	?	?	✓

But if we aim to visualize networks in ways that are clear, simple, informative, replicable and ethical–in alignment with scientific purposes and principles–a single network visualization strategy likely does not equally accomplish those tasks across the different purposes outlined above. In certain cases, analytic network representations likely better serves our purposes. **Table 1** heuristically assesses which of the delineated purposes and/or principles are accomplished across the outlined visualization orientations.

Implementation

My proposal, is that relative to observed (or conceptual) visualizations, modeled network visualizations better accomplish the combined aims of: theoretical implications of parsimonious empirical interpretation, alongside preserving confidentiality and the capacity for reproducibility. While the particularities of the claims in **Table 1** likely leave room for debate, it seems unlikely for a convincing case to be available that *all* (combinations of) aims can equally be satisfied by a single visualization orientation. So, what does model-based network visualization look like?

As with recent resurgence in understanding for the importance of *descriptive* statistics before conducting and presenting any data analyses (Ayala and Koch 2019; Healy and Moody 2014; Wickham et al. 2023), the **first** step is to provide a preliminary account for the broad details of the data to be examined. The full details of what this should entail are likely dependent upon the scientific aims and nature of the data to be investigated (e.g., qualitative or quantitative, closed or open population, etc.), but scholars have recently suggested potentially standardizing these to include a variety of features at the node, tie, and network levels (Neal et al. 2024). Additionally, it is perfectly reasonable to suggest that in some–but as with my proposal here, *not all*–cases, this descriptive task may be improved by a presentation of an observed network visualization.

The **second** step is to conduct whatever forms of analyses are appropriate to the theoretical aims of the research questions and the nature of the analytic dataset. This step necessitates one of the most important

[11] I feel like I want to include a parallel to "writing as thinking" here (Zinsser 1993). Would that be forced?

assertions of the proposal here - “modeled” results come in many forms. While the illustrations below draw from *statistical* models (ergms, in particular), the approach could equally be applied to *any* other form of data analytic results–ranging from similarly statistically oriented models such as SAOMs straight to the primary takeaways from grounded theoretically-oriented ethnographic observations.[12]

Analytic results and interpretations are necessarily reductions of the full-details available in our data (adams 2020). Done well, they draw our attention to the scientifically useful features of the data, lead us to appropriate interpretations, and limit the distractions that keep us from achieving those aims–regardless of the particularities of the case at hand (Healy 2017). The **third** step is therefore to compile an appropriate overview of analytic results in summary form (whether statistical or otherwise).

The **fourth** step is then to convert that summary into a “predicted” network that conforms to the key features of those analytic results. This modeled-network visualization will necessarily differ from any observed network visualization, by eliminating features that are not “modeled” (Schwabish 2021), reducing the prominence of idiosyncratic variation (i.e., identifying “what is this a case of?” (Ragin and Becker 1992; Collins et al. 2024)), and highlighting only the key elements of the findings that warrant theoretical interpretation (Vertesi 2025). The idea here is to provide an empirically-anchored, theoretically-prioritized, visualization that assists in clearly communicating the researchers’ interpretations.

Illustrative Examples

To illustrate this process in practice, I proceed with three examples: (1) a common strategy used in model *building* that illustrates and leverages the implications of the proposed strategy; (2) corollary approaches that leverage modeled results to visualize theoretical takeaways (in non-networks and networks research); and (3) a practical example demonstrating the application to an actual network research question.

Modeling Aid

The first example comes from a classic tutorial approach to modeling social network data (Krivitsky et al. 2026), and draws directly from an implementation that first drove home the utility of this approach for me (Lind 2012). Building statistical models of observed network data can be a complicated process. In the process of developing “well behaving” models, various estimates of goodness of fit can readily assist with interpreting what the latest model estimates are (and are not) accounting for in the data. Many tutorials for developing such models employ visualizations of a single network instance, both as a means to intuit how models work when teaching how to estimate them, and as a means to assist in enhancing model estimation and fit when conducting research (Krivitsky et al. 2026).

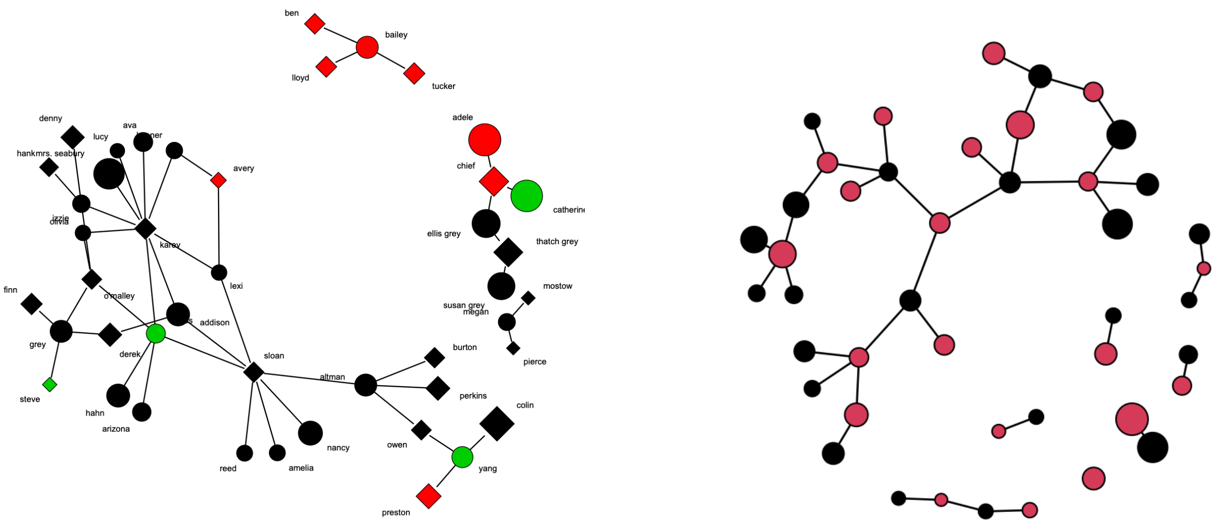
Benjamin Lind (2012) demonstrated this utility incredibly effectively by iteratively updating a model estimation of the “hookup” network among characters in *Grey’s Anatomy* as a means to teach how exponential random graph models work. In **Figure 1**, panel A presents the observed network, while panel B visualizes a single draw from the ergm results, allowing comparison of their intelligibility. First, the modeled visualization only includes those aspects that were incorporated in the model (gender *hetero*phily, and a restriction on isolates[13]), while diminishing the role of idiosyncratic aspects of the data.[14] Second, removing the names does not diminish the ability to observe important features of the network (a general, but not absolute heterosexual partnering pattern).[15] If these were actual data, that carry confidentiality concerns, the unlabeled, and in

[12] As will be noted below, many network data analytic approaches already lean on visualizations–at least in part–to *assist* with interpretation.

[13] Given the way the data were recorded, isolates were not possible in the observed data.

[14] Here, the visual specifically excludes node labels, and nodal attributes of race and age information, because those did not contribute to the observed relational pattern. Furthermore, the “visual” similarity between the general structure of the two graphs–while other, lesser performing models showed obvious divergences (e.g., ones with terms for race or age homophily)–is also confirmed by more formal goodness-of-fit tests.

[15] Moreover, this elides the issues that arise from an often ignored detail. Observed data are necessarily a single characterization of reality, that by invariably includes some level of error. Presenting the “raw” data often overly reifies that particular capture of reality in ways that assume the “realized instance” is the goal of our analyses. Rather, in many cases, the specific instance is only reasonable approximation of our theoretical questions–us removing names avoids drawing our attention towards the idiosyncrasies of the case at hand, and towards the *implications* of the research (Becker 2017).



(a) observed network

(b) modeled network

Figure 1: Two visualizations of *Grey's Anatomy* hookup network. Panel A represents the "observed" data with identities and multiple attributes attached (size=age, color=race, shape=gender). Panel B represents a network generated from the modeled results, including only those exogenous features that are part of the model (node color=gender). This example is adapted from Lind's (2012) ergm tutorial.

being characterized of modeled results these are "representative" nodes and ties, not "capturing" actual relationships, thus also removing any risks of sequential deductive disclosure.[16]

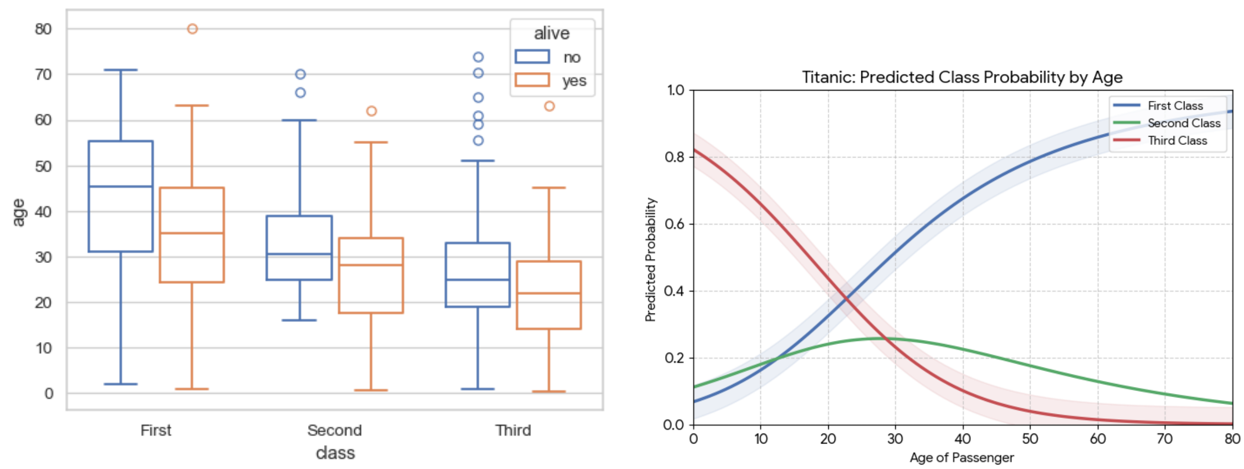
Corollaries

Logistic Regression Let me draw a few parallels here that illustrate how researchers leverage (parts of) the logic of my proposal when *presenting* results rather than within modeling processes. Logistic regression coefficients (just like exponential random graph models) are notoriously hard to make substantive sense of on their own (e.g., significance and magnitude of effects are frequently conflated in results interpretations by writers and readers alike). A common solution to this has been for researchers to: (1) gather, pre-process, then provide a descriptive summary of their data to given an overview sense of what the sample being analyzed actually represents, and how key focal features are distributed in the population; (2) conduct statistical analyses to investigate the researchers' theoretical aims, which are summarized with traditional combinations of betas, standard errors, log-likelihoods, etc. to represent the fit of those claims to empirical estimation, then (3) transform those modeled results into "predicted probabilities" in ways that help interpret what those modeled results "look like" in their implications (Fox and Andersen 2006). To "tell the story" that arises from the model results we don't return to the raw data, provide the individual-level combination of variables for some subset of the population and rely on those case's patterns to convey the implications of our research (in the rare cases that people do, they are often "synthetic" cases, not real ones, e.g., Ayala and Koch 2019).

Likely able to considerably reduce this once I have the example I want to leverage. May be as simple as "plot", model, "predict" and show them in tandem to make the points below.

To do so would not only fail to clearly tell the aggregate story we've learned in the course of our theorization and analysis; it would also violate a number of human subjects protections researchers are obligated to follow. So why in the network world would we need to effectively present the "real" pattern of relationships from our data as a means to interpret and convey our findings in visual form?

[16]For example, given the infrequency of same-sex or inter-racial relationships in the population, even an *unlabeled* version of the graph on the left would allow partners of Derek or "Chief" to likely identify *themselves* in the plot, and thus "discover" their partners having other partners.



(a) Scatterplot of Observed Association **(b)** Predicted Probabilities from Logistic Regression

Figure 2: These are currently stand-ins until I have the example I want to leverage here. This example is adapted from (cite).

Similar *Network* Approaches It is also important to note that the basis of the general idea–if not the complete practice I am recommending–is already in use by researchers in some clusters of social networks. Perhaps the most direct corollary is how blockmodeling results tend to be presented as a capture of role profiles (Doreian et al. 2005). The image matrix and/or reduced graph presentation is a distilled presentation of the overall-role structure in an analyzed graph presented in summary form (Borgatti et al. 2024). A permuted adjacency matrix arranged by density blocks maps that summary onto the full data, but typically for the *observed* adjacency matrix, rather than a modeled representation thereof. Other “coarse graining” visualization strategies of “large” networks take similar parallel approaches.

Another approach to leveraging visualizations to characterize analytic results in theoretically focused ways comes via *network motifs* (Schwarze and Porter 2021). Network motifs seek to identify *repeated* structural patterns across multiple (often, but not exclusively *personal*) subgraphs of networks–and presents those in abstracted “common forms,” rather than presenting multiple parallel observed networks then highlighting such commonalities across the repeated panels. These aims are similar to those in blockmodels, though exclusively considering *local* configurations. However, network motifs–and their visual style of presentation–remain more widely adopted among network science than social network researchers to date (for an exception see Vacca 2020). Other approaches in ego network research similarly draw on stylized representations to visualize analytic implications, rather than reverting to empirical networks to explain findings (especially when doing so would violate confidentiality - McCarty et al. 2019).

Such configurational representations have also been employed to assist interpreting the results inferential statistical models of social network data–by presenting visualizations of micro-patterns rather than coefficient names in results tables (e.g., in stochastic actor based models Snijders and Steglich 2026)

Practical Example

Ideally here, I want data data that are “non-confidential” in any sense to demonstrate why confidentiality concerns should be attuned to in data that are (i.e., I need to be able to show the “revealed” and presumably “unrevealed” versions in tandem to show that the “non-disclosed” version can sometimes be showing more than we think it is).

Here I will again present a “parallel” figure that characterizes the “raw” data, then a “modeled result” that conveys the research takeaway.

Work out the "aims" of visualization a bit more concretely, and elaborate how modeled viz may do this even better than raw data viz can.

I actually think I don't want to use an ERGM example here.}

Quo Vadimus?

Sometimes but not always beneficial. Perhaps better meets the unique combinations of purposes and principles for conveying the theoretical implications of analytic results. Additional benefit of confidentiality.

Hopefully we all can make our continued use of visualizations even more impactful than they've been too date.

I think I want to end with a punchy metaphorical takeaway here - viz is undoubtedly an incredibly powerful tool in the toolkit; unfortunately we've come to only see it as a hammer. Or to awssume a hammer's the tool we need. When at times the needs are clearly calling for a tool with different qualitites.

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